

Performance Prediction and Optimization of Supercapacitors Through Data-Driven Analysis

Project Review

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Seminar Outline

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Introduction

- **Context:** Supercapacitors are advanced energy storage devices that provide high power density and fast charge-discharge capabilities.
- **Current Challenges:** There is a lack of predictive models capable of accurately determining the optimal combination of substrate, electrode material, electrolyte type, and electrolyte concentration and predict capacitance.
- **Solution:** Applying machine learning to optimize supercapacitor material selection and predict capacitance accurately.
- **Impact:** Enhancing energy storage efficiency in applications such as electric vehicles and renewable energy systems.



Motivation

● Why Supercapacitors?

- High energy and power density compared to traditional capacitors.
- Longer cycle life and rapid charging make them ideal for various applications.

● Why Machine Learning?

- Reduces the time required for experimental trials by predicting material performance.
- Can Identify optimal material combinations required to efficiently design the supercapacitor.



Problem Statement

- **Current Issues:**

- Lack of a structured predictive approach for capacitance optimization.
- Experimentally determining the best material combination is time-consuming and costly.

- **Objective:**

- Develop an ML-based approach to predict capacitance based on material properties.
- Identifying the best material combinations for a specific target in supercapacitor design.



Selected Literature Survey

● Existing Solutions:

- Research on supercapacitors has primarily focused on experimental methods.
- Machine learning has been introduced as an alternative approach but remains underutilized in this domain.
- Studies have demonstrated that ML models can predict capacitance with reasonable accuracy, reducing the dependency on physical trials.



Selected Literature Survey

- **Our Contribution:**

- **Capacitance Prediction Model:** Developed and optimized a machine learning model to accurately predict the capacitance of supercapacitor based on a variety of input factors (e.g., specific surface area, pore width, pore volume etc.).
- **Material Combination Optimization:** Proposed a novel approach for predicting the best material combinations (substrate, electrode, electrolyte, electrolyte conc.) for supercapacitors aimed at achieving specific performance targets (e.g. energy density, power density and areal capacitance).



Methodology

- **Data Collection and Preprocessing:**

- Extracting capacitance-related data from published research.
- Use of Gaussian noise to expand the dataset for more robust training.
- Handling missing values, standardizing numerical features, and performing correlation analysis.

- **Feature Selection:**

- Identifying key parameters influencing capacitance, such as specific surface area, pore width, and pore volume.

- **ML Model Selection:**

- Evaluating various algorithms to determine the most suitable model for prediction.



Dataset Overview I

Table 1: Dataset for Capacitance Prediction

No.	PW (V)	SSA	PV	PS	ID/IG	N%	O%	Capacitance (F/g)
1	0.8	0.68	0.72	4.00	1.62	0.00	10.75	100.00
2	0.8	1.38	1.20	1.00	1.30	0.00	9.00	160.00
3	1.0	0.38	0.51	5.40	1.15	1.53	3.10	227.00
4	1.0	2.01	1.60	0.55	1.12	0.10	6.27	319.00
5	1.0	0.34	1.60	0.62	1.38	0.00	1.54	7.86



Dataset Overview II

Table 2: Dataset for Material Prediction

Electrode	Electrolyte	Substrate	Electrolyte Conc. (g/L)	Areal Capacitance (mF/cm ²)	Current (mA/cm ²)	Scan Rate (mV/s)	Energy Density (uWh/cm ²)	Power Density (mW/cm ²)
LIG	PVA-H ₃ PO ₄	Kapton PI	85.00	19.47	0.04	2	3.38	0.199
2% CNTs	PVA/H ₂ SO ₄	Kapton PI	125.00	51.975	0.20	2	6.50	0.219
5% CNTs	PVA/H ₂ SO ₄	Kapton PI	125.00	28.375	0.20	2	5.01	0.199
GO-MnO ₂	PVA-H ₃ PO ₄	Kapton PI	5.00	4.02	0.50	10	1286.4	–
LIG-MXene	H ₃ PO ₄ /PVA	Kapton PI	–	2.19	0.10	10	–	–
LIG-C	PVA/H ₂ SO ₄	Parylene-C	85.87	1.66	0.50	10	0.19	0.00459
B-doped LIG	PVA/H ₂ SO ₄	Kapton PI	163.49	16.5	0.05	10	–	–
RuO ₂ /Au	PVA/H ₂ SO ₄	Kapton PI	90.90	27.0	2.70	100	3.1	2.3
NiCo ₂ S ₄	Co-Ni-Thiourea	Kapton PI	228.36	30.4	0.75	5	16.9	0.368
d-LIG-D	PVA-H ₂ SO ₄	Kapton PI	98.08	19.5	0.05	2	2.7	0.025
LIG	PVA/H ₂ SO ₄	Whatman Paper	87.62	4.6	0.50	10	0.3	0.0045



Model Development

- **Algorithms Used:**

- Capacitance Prediction: Decision Tree, XG Boost, Random Forest, SVR, MLP, LR, Stacked Model
- Material Prediction: Decision Tree, XG Boost, Random Forest, Gradient Boosting

- **Evaluation Metrics:**

- R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

- **Optimization Techniques:**

- Hyperparameter tuning using Grid Search and Bayesian Optimization.
- Training (80%) and testing (20%) sets for model evaluation.



Correlation Heatmap

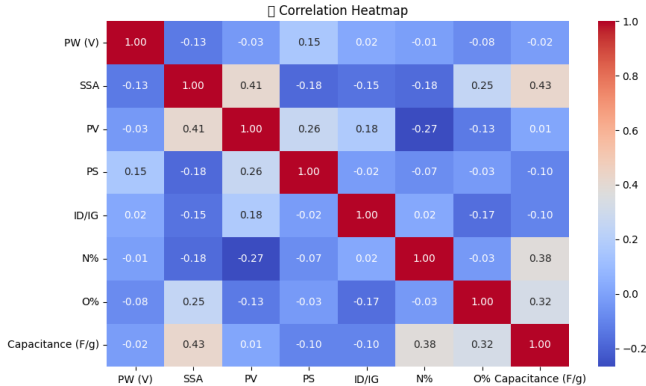


Figure 1: Correlation Heatmap



Capacitance Prediction Results

Table 3: Performance of Different ML Models

Model	RMSE	MAE	R ²	Accuracy (%)
Stacked Model	25.53	16.86	0.95	95.36
Random Forest	26.96	17.35	0.95	94.82
XGBoost	29.58	17.98	0.94	93.77
Decision Tree	53.68	39.18	0.79	79.47
LR	67.81	51.32	0.67	67.25
SVR	73.91	50.72	0.61	61.09
MLP	73.96	56.60	0.61	61.04



Material Combination Prediction Results

Which value do you want to use as input?

1. Areal Capacitance (mF/cm²)
2. Energy Density (uWh/cm²)
3. Power Density (mW/cm²)

Enter option (1/2/3): 2

You selected: Energy Density (uWh/cm²)

Allowed range: Min = 0.003, Max = 17.476

Enter your desired value for Energy Density (uWh/cm²): 0.4

Do you want to use only LIG electrodes? (yes/no): no

 Reverse Model Evaluation (80-20 split):

Model	MAE	R (Correlation)	R ²	Accuracy (%)
Decision Tree	4.823	0.998	0.996	99.61
Random Forest	4.137	0.998	0.997	99.68
Gradient Boosting	4.375	0.998	0.997	99.67
XGBoost	4.735	0.998	0.996	99.65

☒ Best model selected: Random Forest

☒ No exact match found in dataset. Proceeding with prediction...

 Predicted material combinations to achieve your target value:

Electrode	Electrolyte	Substrate	Predicted Conc. (g/l)
LIG	PVA/H2SO4	Cork	81.55
LIG-C	PVA/H2SO4	parylene-C	82.57
LIG	PVA/H2SO4	Whatman chromatography paper	82.88
LIG	PVA/H3PO4	Kapton PI	86.18
RuO2/Au	PVA/H2SO4	Kapton PI	89.50



Conclusion and Future Scope

- **Conclusion:**

- Successfully developed machine learning models for both capacitance prediction and material combination optimization in supercapacitor systems.
- Demonstrated that data-driven approaches can accurately predict areal capacitance and recommend high-performance material configurations
- Manuscript currently in preparation for submission

- **Future Work:**

- Expand the dataset with more experimental entries to further enhance model robustness.
- Validate the predicted material combinations through experimental fabrication and electrochemical testing.



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Thank You

